- **Biasing Diodes:** Diodes D1–D4 are placed between the Gates of Q1 and Q2 to create a fixed voltage drop.
- **Quiescent Current:** Resistors R19 and R20 ensure these diodes remain forward-biased. This keeps the MOSFET Gates held closer to their activation voltages.
- **Linearity:** By reducing the "swing" the op-amp must perform to switch between transistors, the system achieves a smoother transition at the zero-crossing point, significantly improving signal integrity for the Function Generator.

3.1.5 Stability and Compensation

Because MOSFET gates possess innate capacitance, they interact with the output resistance of the op-amp and R21 to form an unintentional RC low-pass filter. This introduces a phase lag in the negative feedback loop.

- **The Oscillation Risk:** At high frequencies, this phase lag can cause the feedback to become positive, resulting in the 12 V rail oscillating rather than stabilizing.
- **Compensation Logic:** Capacitor C1 was introduced as a compensation element. It provides a high-frequency feedback path that bypasses the lag-inducing MOSFET gates.
- **Result:** This ensures an adequate phase margin, forcing the virtual ground to remain stable at exactly 12.0 V even when subjected to high-frequency load changes from the SPWM generation.

A Miller compensation capacitor reduced high-frequency gain, ensuring stable operation.

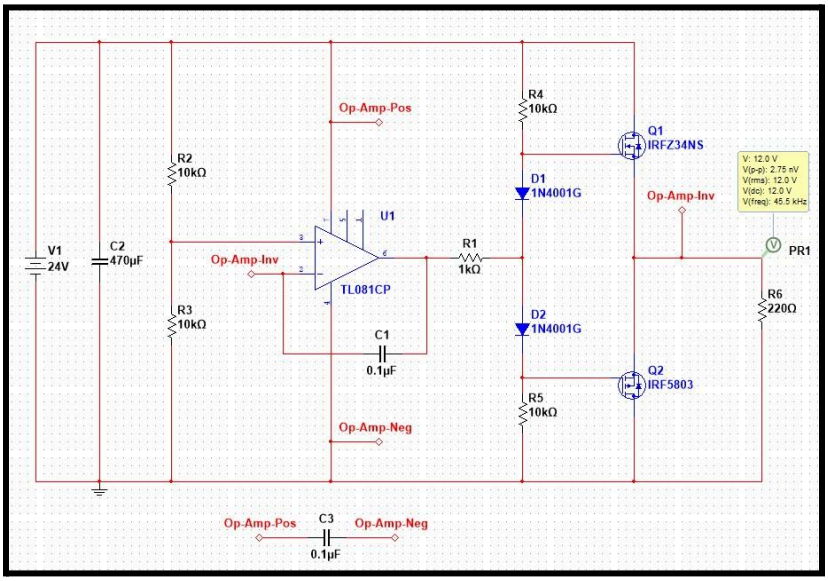


Figure 3.1 – Power Supply Design Schematic